Name: D S LAKSHMI NARAYANA

Register No.: 192512371

Course Name: SIGNALS AND SYSTEM

Course Code: ECA 0304

EXPERIMENT – 13: SAMPLING THEOREM VERIFICATION AND SAMPLING-FREQUENCY SELECTION

OBJECTIVES

• Generate a continuous-time sinusoidal signal and sample it at sub-Nyquist, Nyquist and super-Nyquist rates.

• Verify the Sampling Theorem for a band-limited signal of maximum frequency fm.

• Reconstruct the original signal using ideal sinc interpolation and quantify reconstruction error.

• Identify an appropriate sampling frequency for a given signal in data-acquisition system design.

Experimental Procedure

1. Generate $x(t)=\cos(2\pi \cdot 50t)$ using a fine time grid with dt=10 μs.
2. Sample x(t) at fs = 0.5fm, fm, 2fm, 5fm and 10fm.
3. Plot the reference waveform with the sampled points overlaid.
4. Reconstruct the signal using sinc interpolation.

Recorded

5. Compute and tabulate the mean reconstruction error against fs/fm.
6. Identify the sampling rates giving the recorded reconstruction performance.

MATLAB CODING

```
clc; clear; close all;
fm = 50;              % message (max) signal frequency, Hz
T  = 0.06;            % observation window, s
dt = 1e-5;            % fine grid to emulate CT signal
t  = 0:dt:T;
x  = cos(2*pi*fm*t);   % reference continuous-time signal
fs_list = [0.5*fm, fm, 2*fm, 5*fm, 10*fm];
labels  = {'fs = 0.5 fm (aliased)','fs = fm (under-sampled)', ...
'fs = 2 fm (Nyquist rate)','fs = 5 fm (over-sampled)','fs = 10 fm (over-sampled)'};
```

```
figure('Name','Sampling at Different Rates');
for k = 1:length(fs_list)
    fs = fs_list(k);
    n  = 0:1/fs:T;
    xn = cos(2*pi*fm*n);
    subplot(length(fs_list),1,k);
    plot(t,x,'b-','LineWidth',1); hold on;
    stem(n,xn,'r','filled');
    title(labels{k}); xlabel('t (s)'); ylabel('x(t)');
end
%% Reconstruction using ideal sinc interpolation + error vs sampling rate
fprintf('Nyquist rate (2*fm) = %.2f Hz\n', 2*fm);
err = zeros(size(fs_list));
for k = 1:length(fs_list)
    fs = fs_list(k);
    n  = 0:1/fs:T;
    xn = cos(2*pi*fm*n);
    x_recon = zeros(size(t));
    for i = 1:length(n)
        x_recon = x_recon + xn(i)*sinc(fs*(t - n(i)));
    end
    err(k) = mean(abs(x - x_recon));
    fprintf('fs = %6.1f Hz (fs/fm = %4.1f) -> mean reconstruction error = %.5f\n', fs, fs/fm, err(k));
end
figure('Name','Reconstruction at fs = 10 fm');
fs = 10*fm; n = 0:1/fs:T; xn = cos(2*pi*fm*n);
x_recon = zeros(size(t));
for i = 1:length(n)
    x_recon = x_recon + xn(i)*sinc(fs*(t - n(i)));
end
plot(t,x,'b','LineWidth',1.5); hold on;
plot(t,x_recon,'r--','LineWidth',1.2);
legend('Original x(t)','Reconstructed x(t)');
title('Sinc-Interpolated Reconstruction'); xlabel('t (s)'); ylabel('Amplitude'); grid on;
```

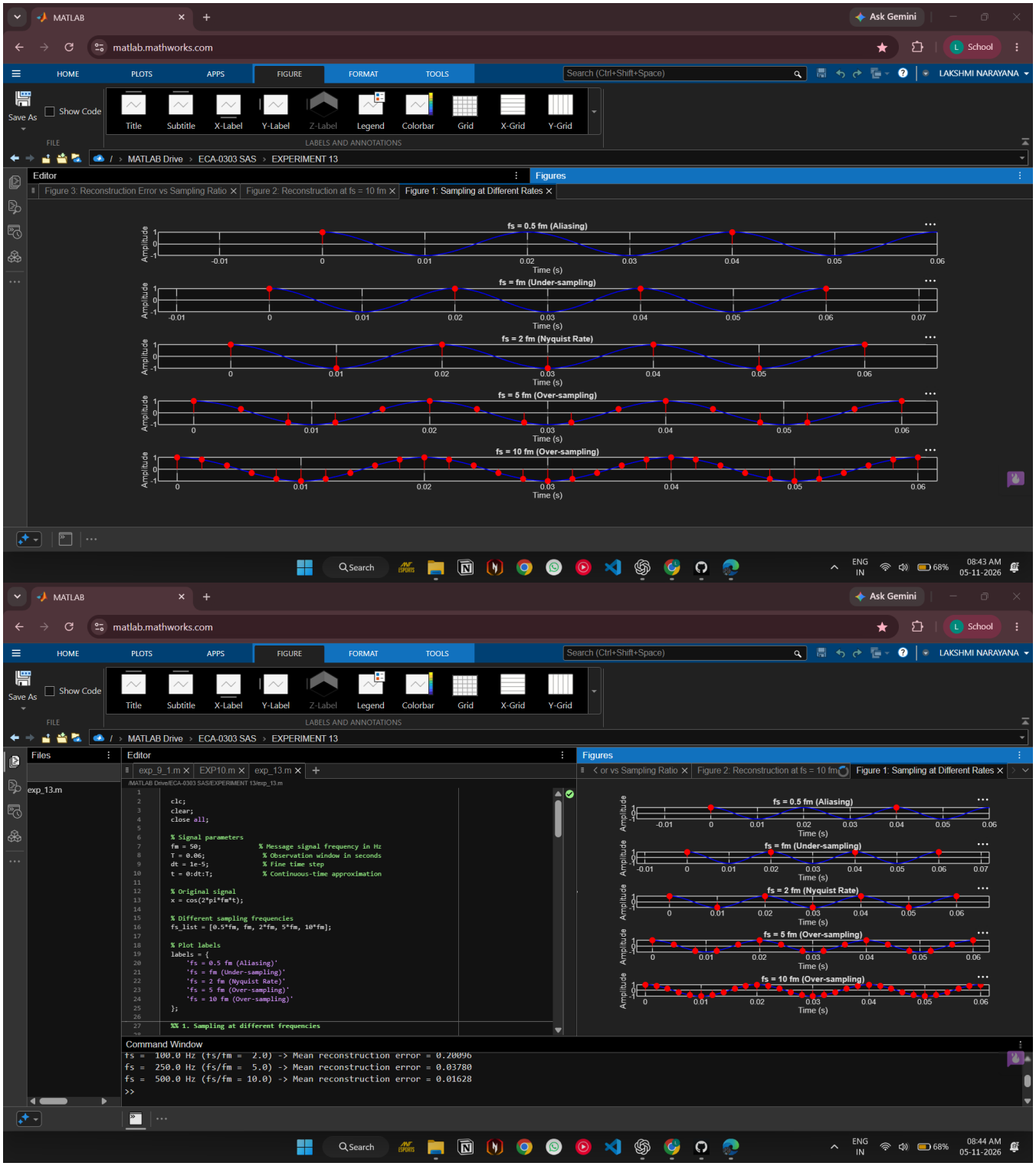
figure('Name','Reconstruction Error vs Sampling Ratio');

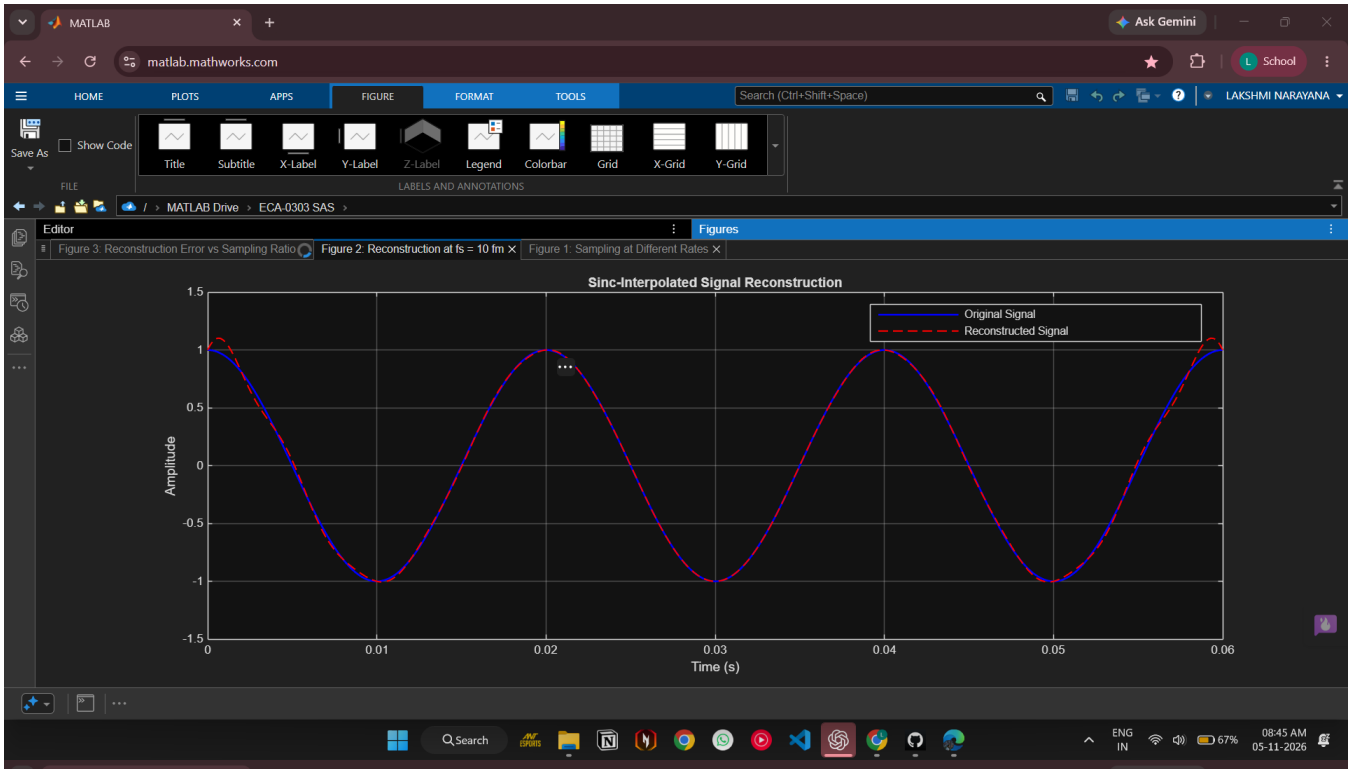
semilogy(fs_list/fm, err, '-o','LineWidth',1.5); grid on;

xlabel('f_s / f_m'); ylabel('Mean reconstruction error (log scale)');

title('Reconstruction Error vs Sampling-to-Signal Frequency Ratio');

Input and Output Graphs




Sinc-Interpolated Signal Reconstruction
Original Signal
Reconstructed Signal
Amplitude
Time (s)
1.5
1
0.5
0
-0.5
-1
-1.5
0
0.01
0.02
0.03
0.04
0.05
0.06


Reconstruction Error vs Sampling Frequency Ratio
Mean Absolute Reconstruction Error
Sampling Ratio (fs/fm)
10^0
10^-1
10^-2
0
1
2
3
4
5
6
7
8
9
10